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Special Collection:
Advanced machine learning in solid earth geoscience

Key Points:

- PyQuake3D is a new open-source tool for simulating earthquake and fault slip behavior in three dimensions
- PyQuake3D models both slow fault motion and fast ruptures while remaining efficient and scalable algorithm
- PyQuake3D enables modeling of the full seismic cycle and assessment of long-term seismic hazard on complex fault systems

Supporting Information:

Supporting Information may be found in the online version of this article.

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PyQuake3D: A Python Tool for 3-D Earthquake Sequence Simulations of Seismic and Aseismic Slip

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Abstract Simulating the full spectrum of fault slip behavior—from slow aseismic creep to dynamic earthquake rupture—is essential for advancing our understanding of fault mechanics and long-term seismic hazard. However, few open-source tools provide efficient three-dimensional modeling of these processes with the flexibility needed for research across scales. Here we introduce PyQuake3D, an open-source Python package for simulating fault slip evolution using quasi-dynamic approximations. The code solves elastic interactions via the boundary element method and incorporates hierarchical matrix compression to reduce memory usage and computational cost. PyQuake3D is designed for scalability and portability, supporting both distributed-memory parallelism (MPI) and optional Graphics Processing Unit (GPU) acceleration. We validate the code against established sequences of earthquakes and aseismic slip (SEAS) benchmarks and illustrate its ability to capture key phases of the seismic cycle, including nucleation, rupture propagation, arrest, interseismic stress accumulation, and postseismic relaxation. Applications include simulations of faults with spatially heterogeneous friction, a geologically constrained model of the East Anatolian Fault Zone, and a laboratory-inspired triaxial experiment. These examples demonstrate the ability of PyQuake3D to resolve the spatiotemporal complexity of fault slip, including the interplay of geometry, frictional properties, and loading. By combining scalable numerical techniques with accessible software design, PyQuake3D provides a robust and extensible platform for modeling fault slip dynamics across a wide range of geophysical scenarios.

Plain Language Summary Understanding how faults accumulate stress and release it through slow or rapid slip is essential for assessing earthquake hazard. Numerical simulations are powerful tools to study these processes, but existing codes are often limited in scope or computational efficiency. This study introduces PyQuake3D, an open-source software package designed to simulate long-term fault slip behavior in three dimensions. The code efficiently models both slow aseismic slip and fast dynamic rupture using a boundary element method combined with advanced numerical techniques. It is scalable on both high-performance computing clusters and standard workstations. PyQuake3D is validated against established earthquake sequence benchmarks and used to explore how fault geometry and frictional properties influence earthquake nucleation, rupture, and postseismic deformation. Additional simulations demonstrate the applicability of the tool to realistic fault systems worldwide. By making these capabilities widely accessible, PyQuake3D provides a new platform for advancing our understanding of the earthquake cycle and for improving long-term seismic hazard models.

1. Introduction

Faults accommodate tectonic deformation through a wide spectrum of slip behaviors, from slow aseismic creep to dynamic earthquake rupture. These phenomena span orders of magnitude in space and time—ranging from seconds-long ruptures to interseismic strain accumulation over decades (e.g., Ben-Zion, 2008; Bürgmann, 2018; Masson et al., 2005; Schwartz & Rokosky, 2007). Rather than forming distinct categories, seismic and aseismic slip evolve along a continuum governed by the interplay of frictional rheology, fault geometry, material heterogeneity, and pore fluid pressure (e.g., Barbot, 2023a; Cebry et al., 2022; Leeman et al., 2016; Y. Liu & Rice, 2005; Romanet et al., 2018; Sathiakumar et al., 2020). Capturing this continuum is essential not only for advancing our understanding of fault mechanics but also for improving long-term seismic hazard assessments (Abdelmeguid & Elbanna, 2022; Avouac, 2015; L. Huang et al., 2025). Although advances in geodetic and seismological observations have greatly improved surface monitoring of fault slip, they cannot directly probe the

10.1029/2025JH000871

evolving physical conditions at depth (Elliott et al., 2016). Data-driven methods, including machine learning, offer powerful tools to identify patterns in seismic and geodetic time series (Bergen et al., 2019; Kong et al., 2019), but they remain constrained by data coverage and resolution. In this context, physics-based models play a critical role, as they allow controlled testing of hypotheses and enable exploration of mechanisms that remain otherwise inaccessible (Barbot et al., 2012; Kaneko et al., 2010).

A key advance in modeling fault slip has been the development of seismic and aseismic sequence simulations. These models incorporate laboratory-derived constitutive laws to describe the evolution of fault strength, particularly the rate-and-state friction (RSF) framework (Dieterich, 1979; Ruina, 1983). RSF captures the dependence of friction on sliding velocity and the history of contact, enabling simulation of a broad range of fault behaviors—including aseismic creep, earthquake nucleation, and dynamic rupture—within quasi-dynamic formulations (Dal Zilio, Hegyi et al., 2022; Dal Zilio, Lapusta et al., 2022; Jiang et al., 2022; Lapusta & Liu, 2009; Rice, 1993; Sathiakumar & Barbot, 2021; Thomas et al., 2014). The ability of RSF models to represent both velocity-strengthening and velocity-weakening behaviors has led to major insights into fault stability across the seismic cycle (Marone, 1998; Scholz, 1998). However, faults are not governed by frictional laws alone. They are complex structures that influence both the mechanical behavior and hydraulic connectivity of the surrounding rock mass (Faulkner et al., 2010). In this broader context, coupled mechanisms—such as the interaction between slip, stress, and fluid pressure—play a critical role in regulating the stability of fault slip and determining how elastic strain energy is partitioned into frictional dissipation, fracturing, and radiated seismic energy (Cocco et al., 2023). Capturing these multiscale energy dissipation pathways is essential for understanding rupture dynamics and the physical diversity of earthquake behavior (Kammer et al., 2024).

Despite these advances, many existing modeling tools remain difficult to use or extend. Most open-source simulators are written in low-level compiled languages such as Fortran, C, or C++ (e.g., Barbot, 2023b; D. Liu et al., 2020; Luo et al., 2017; Ozawa et al., 2023; Romanet & Ozawa, 2021). Although efficient, these codes present steep learning curves and limit rapid experimentation, especially for non-specialist users. In contrast, Python has emerged as a powerful language for scientific computing due to its readable syntax, extensive library support, and growing capabilities for high-performance computing, including GPU acceleration and distributed-memory parallelism (Harris et al., 2020; Virtanen et al., 2020).

Here we introduce PyQuake3D: a fully open-source, Python-based simulator for quasi-dynamic earthquake cycle modeling in three dimensions. PyQuake3D leverages the Boundary Element Method (BEM) to compute elastic stress interactions and supports a wide range of fault geometries, frictional properties, and slip behaviors. To balance accessibility and computational performance, the code includes both GPU-accelerated dense algebra and a CPU-based implementation using hierarchical matrices and MPI-based parallelism. PyQuake3D is designed to be modular and extensible, allowing researchers to test new physical assumptions, integrate custom boundary conditions, and simulate complex fault networks across spatial scales. We validate PyQuake3D against community benchmarks and illustrate its capabilities through a suite of applications, including simulations of heterogeneous frictional faults, multi-segment fault systems, and laboratory-scale stick-slip experiments. By combining physical rigor with modern computational design, PyQuake3D aims to bridge the gap between theory and application, enabling broader engagement in earthquake modeling and seismic hazard research.

This paper is organized as follows. Section 2 outlines the overall computational architecture and the trade-offs between CPU- and GPU-based solvers. Section 3 presents the governing equations and physical formulation. Section 4 validates the simulator against a standard benchmark problem. Section 5 demonstrates the flexibility of the code through several geophysical and experimental applications. Finally, Section 6 synthesizes the technical advances and outlines future directions for development.

2. Methods

2.1. Computational Framework

The computational design of PyQuake3D is illustrated in Figure 1 and reflects a modular, performance-optimized architecture tailored for both small- and large-scale earthquake simulations using three-dimensional boundary element methods. The framework consists of three key stages—input, simulation engine, and output—and supports both CPU- and GPU-based parallel computing for enhanced scalability and flexibility. The input phase involves defining the fault geometry and initial conditions. PyQuake3D supports unstructured triangular

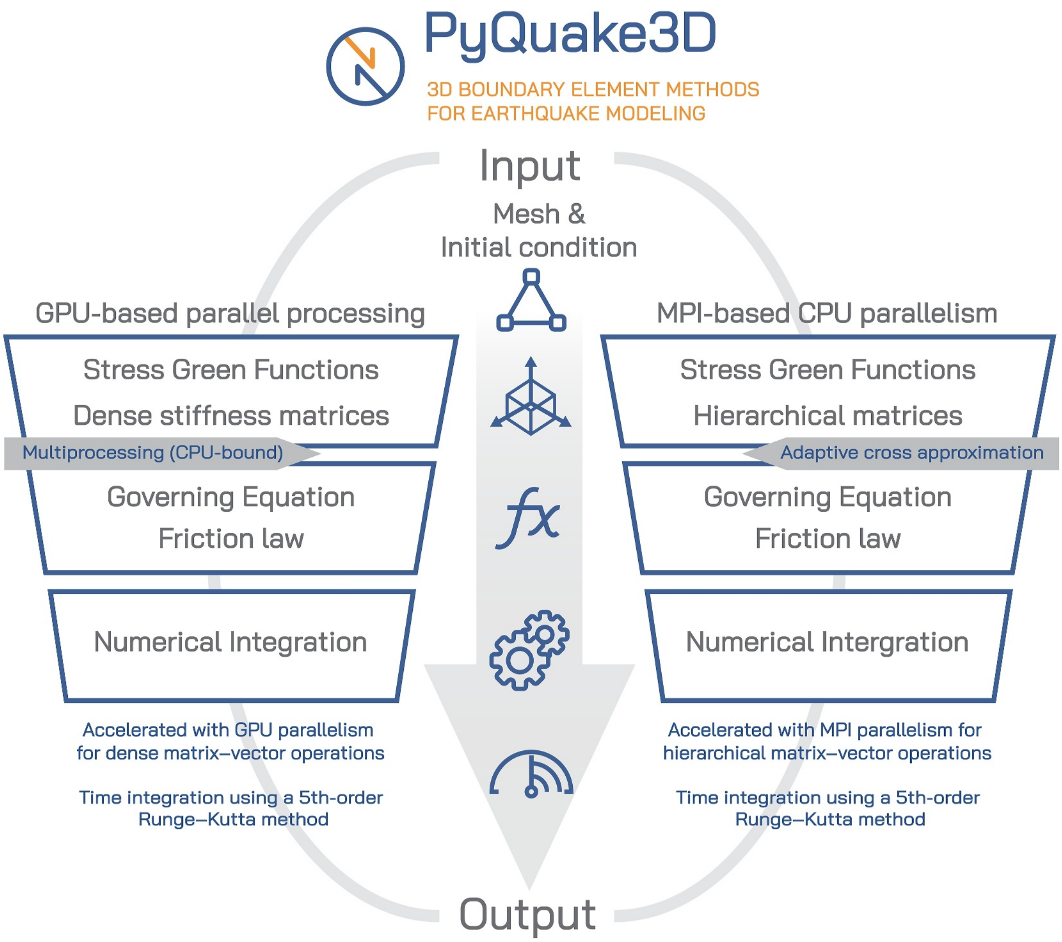

Figure 1. Overview of the PyQuake3D computational framework. The workflow includes input setup, solver implementation on GPU or CPU architectures, and output diagnostics. The GPU version uses dense matrix-vector operations accelerated with CUDA, whereas the CPU version applies hierarchical matrix compression and MPI-based parallelism.

surface meshes provided in the `.msh` format compatible with `Gmsh`, facilitating simulations on geometrically complex, multi-fault systems. The framework accommodates faults with features such as stepovers, curvature, and segment discontinuities, embedded within either a full-space or half-space elastic medium. This geometric flexibility enables the modeling of a broad range of tectonic settings with realistic fault configurations.

At the core of the implementation lies the formulation of the governing equations and the frictional behavior of the fault interface. The fault system is discretized into elements, each governed by a set of coupled ordinary differential equations (ODEs) describing quasi-dynamic rupture evolution. The interaction between elements is mediated through precomputed stress Green's functions, which define how slip on a source element induces stress changes on all other receiver elements. These stress interactions give rise to dense stiffness matrices in the GPU version and hierarchical matrices in the CPU version.

In the GPU-based implementation, dense stiffness matrices are constructed by explicitly evaluating stress interactions between all pairs of source and receiver elements. Matrix-vector multiplications, which represent the primary computational bottleneck, are accelerated using `CuPy` and GPU-optimized linear algebra routines. Independent kernel evaluations are further parallelized using Python's `ProcessPoolExecutor`. Time integration of the resulting system of ordinary differential equations is performed using a fifth-order Dormand-Prince Runge-Kutta scheme with adaptive time stepping. In contrast, the CPU-based implementation leverages a hierarchical matrix (H-matrix) representation to reduce memory usage and computational cost. This is achieved via Adaptive Cross Approximation (ACA), which approximates low-rank off-diagonal blocks of the interaction matrix without constructing the full matrix explicitly. Parallelism is implemented using `mpi4py`, enabling

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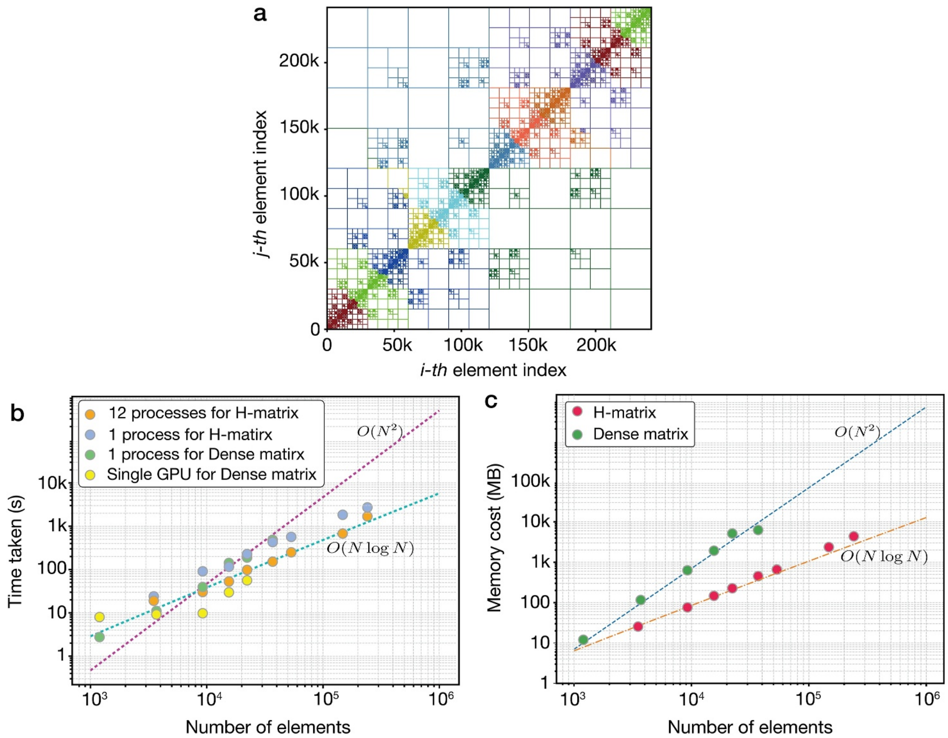

Figure 2. Performance and memory scaling of the PyQuake3D solver. (a) Block structure of the stiffness matrix using hierarchical matrix (H-matrix) compression for a 3D fault mesh with over 240,000 elements. Each colored block represents a submatrix, with far-field blocks stored in low-rank form and near-field blocks stored fully. (b) Runtime comparison of the BP5-QD model across varying cell counts, executed 500 iterations, using different solver backends and hardware configurations. The H-matrix approach scales as $O(N \log N)$, whereas the dense matrix approach scales as $O(N^2)$. GPU acceleration offers the lowest runtime for moderate problem sizes. (c) Memory cost as a function of number of elements. H-matrix compression significantly reduces memory usage compared to dense matrices, particularly for large-scale problems.

distributed-memory computation of both Green's functions and matrix-vector products across multiple processors. This version is particularly well suited for high-resolution simulations involving more than 40,000 fault elements.

The simulation engine tracks the evolution of slip, stress, and other rupture metrics over time and stores outputs in an accessible format. Simulation results can be queried directly through Python, facilitating seamless post-processing, visualization, and coupling with external tools. Overall, the PyQuake3D framework balances numerical rigor, computational efficiency, and user accessibility. Its dual implementation strategy—using either GPU-accelerated dense algebra or CPU-based H-matrix solvers—provides flexibility for running both exploratory simulations on local workstations and large-scale rupture models on high-performance computing clusters. Figure 1 provides a graphical overview of the framework and its parallel computing options.

2.2. Hierarchical Matrix and GPU/CPU Performance Trade-Offs

To enable scalable matrix-vector operations in large-scale simulations, we developed a custom hierarchical matrix (H-matrix) implementation within PyQuake3D. Unlike dense representations that scale as $O(N^2)$ in both time and memory, H-matrices exploit the low-rank structure of long-range (far-field) interactions to reduce overall complexity to approximately $O(N \log N)$, where N is the number of fault elements (Ambikasaran et al., 2019; Börm et al., 2003). Our H-matrix implementation is specifically tailored for boundary element stiffness matrices arising from fault interaction kernels, and it is integrated into the parallel CPU solver of PyQuake3D.

The H-matrix framework partitions the full stiffness matrix into a hierarchy of sub-blocks classified as either near-field or far-field (Figure 2a), based on a geometric admissibility criterion. Near-field blocks, which correspond to closely interacting fault elements, are stored densely to preserve numerical accuracy. Far-field blocks, which capture long-range interactions, are approximated in low-rank form using Adaptive Cross Approximation (ACA) (Bebendorf, 2000; Rjasanow & Steinbach, 2007). ACA constructs the low-rank factorization iteratively by sampling selected rows and columns, avoiding the need to assemble the full dense block and significantly reducing computational cost. Further implementation details and algorithmic structure are provided in Supporting Information S1 (Text S1, Figure S1).

To support distributed-memory parallelism, the H-matrix is assembled and applied using `mpi4py`. Each MPI process handles a subset of matrix blocks and independently computes local contributions to the matrix-vector product. Low-rank approximations are constructed locally, minimizing inter-process communication. At each time step, the matrix-vector products from each process are aggregated using `MPI_Reduce`, ensuring scalability on high-performance computing clusters. Figures 2b and 2c compare the runtime and memory usage of the H-matrix and dense matrix approaches over 500 iterations of the BP5-QD benchmark (Section 4). The H-matrix method exhibits near $O(N \log N)$ scaling, whereas the dense matrix approach scales approximately as $O(N^2)$. For smaller problem sizes, dense matrix computations on the CPU utilize only a portion of the available resources, leading to deviations from the expected $O(N^2)$ behavior. As the model size increases, a larger share of computational resources is allocated to matrix-vector multiplication. In this regime, the underlying parallelized implementation of NumPy achieves performance that more closely reflects the theoretical $O(N^2)$ scaling.

We evaluated GPU performance using a single NVIDIA GeForce RTX 3090, which is limited by 24 GB of video memory. This constraint restricted testing to models with fewer than 30,000 elements. GPU performance did not scale linearly with model size due to two primary factors: for small models, overhead from thread scheduling and memory access led to suboptimal performance compared to CPU-based computations; for larger models, limited GPU memory necessitated fallback to CPU assistance, which introduced performance degradation-though still yielding faster runtimes than MPI-parallelized H-matrix CPU implementations.

Dense matrix-based calculations on both GPU and CPU are fundamentally constrained by their explicit memory requirements and $O(N^2)$ scaling behavior, which severely limits applicability to large-scale meshes (Figure 2c). In contrast, the H-matrix formulation offers a more favorable trade-off between speed and memory efficiency. For models exceeding 30,000 fault elements, the H-matrix CPU implementation maintains near-linear scaling and enables high-resolution simulations that are otherwise infeasible with dense matrix methods due to memory limitations.

Overall, our custom H-matrix implementation—based on hierarchical compression, ACA-based low-rank approximation, and distributed parallelism—extends the capability of PyQuake3D to model realistic, high-resolution fault systems efficiently. This approach supports a wide range of model sizes and geometries, offering a robust computational foundation for long-term earthquake cycle simulations.

3. Governing Equations for Quasi-Dynamic Earthquake Cycle Simulation

To simulate fault slip evolution under quasi-dynamic conditions, we formulate a system of boundary integral equations embedded in a homogeneous elastic half-space. A uniform tectonic loading rate is imposed across the fault interface, and slip-induced elastic stress transfer is computed using precomputed Green's functions. The governing expressions for the shear and normal stress at each fault element are given in Equations 1 and 2, incorporating the quasi-dynamic approximation with radiation damping (Rice, 1993):

$$\tau_i = \tau_0 - \sum_{j=1}^{n} k_{ij}^{s}(u_j - V_{pl}\, t) - \frac{\mu}{2c_s}\frac{\partial u_i}{\partial t} \tag{1}$$

$$\sigma_i = \sigma_0 + \sum_{j=1}^{n} k_{ij}^{N}(u_j - V_{pl}\, t) \tag{2}$$

where V_{pl} is the imposed tectonic slip rate, μ is the shear modulus, c_s is the shear wave speed, and u_j is the slip at the j-th element. The kernels k_{ij}^{s} and k_{ij}^{N} represent the shear and normal stiffness matrices, respectively. The last

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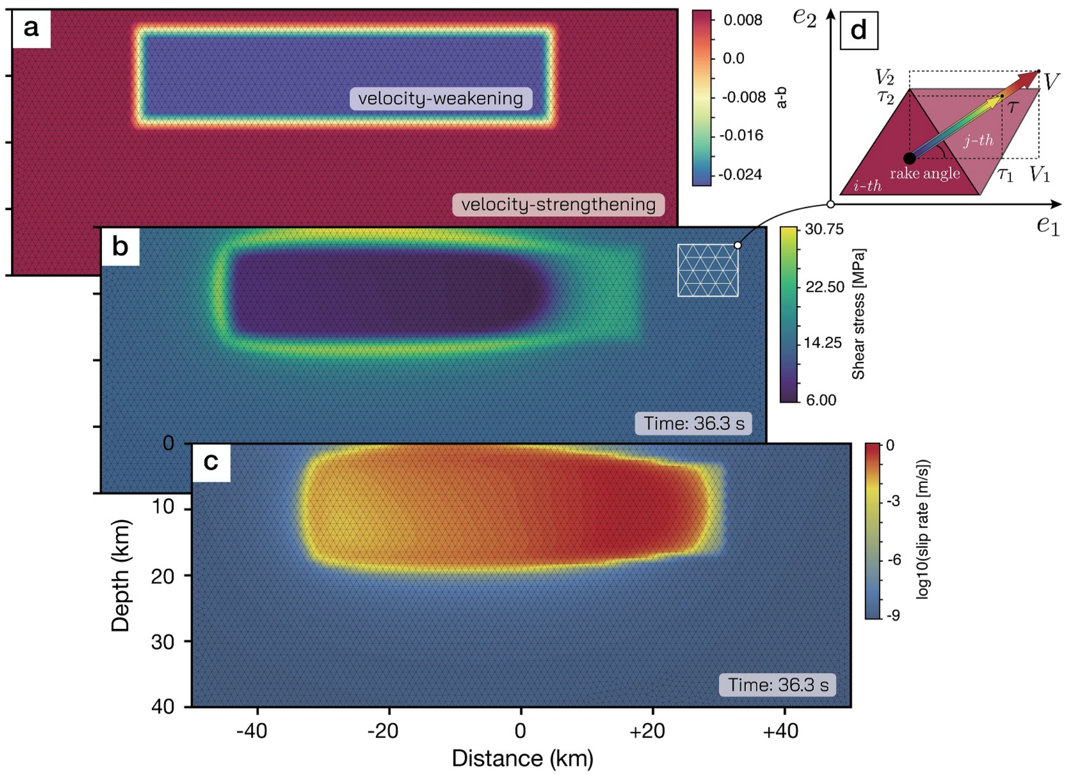

Figure 3. Model configuration for the quasi-dynamic earthquake cycle simulation. (a) Spatial distribution of the frictional parameter $a-b$, highlighting velocity-weakening and velocity-strengthening regions. (b) Shear stress field on the fault at a selected time step. (c) Slip rate field (log scale) at a selected time step. (d) Schematic of the local slip vector and rake angle definition for triangular elements.

term in Equation 1 captures radiation damping and approximates inertial effects. The τ_0 and σ_0 are the tectonic loading for initial shear stress and effective normal stress on the fault surface.

The frictional strength is modeled using a regularized rate-and-state framework with the aging law (Dieterich, 1979; Ruina, 1983):

$$f(V,\theta) = a\sinh^{-1}\left[\frac{V}{2V_0}\exp\left(\frac{f_0 + b\ln\left(\frac{V_0\theta}{d_c}\right)}{a}\right)\right] \tag{3}$$

$$\frac{d\theta}{dt} = 1 - \frac{V\theta}{d_c} \tag{4}$$

where d_c is the characteristic slip distance, V_0 the reference slip rate, f_0 the reference friction coefficient, and θ the state variable. Parameters a and b represent the direct and evolution effects of frictional resistance. The typical model setup includes a fault discretized with triangular elements, incorporating spatial heterogeneity in frictional properties defined by the parameter $a-b$, which distinguishes between velocity-weakening and velocity-strengthening behavior (Figure 3). As shown in Figure 3a, a central velocity-weakening patch is embedded within a broader velocity-strengthening domain, allowing for spontaneous rupture nucleation, propagation, and arrest. This frictional structure is critical for capturing the full spectrum of slip behavior governed by the rate-and-state framework.

To integrate these laws with the elastic interaction, the full system is reformulated into a set of first-order ordinary differential equations (ODEs) using chain rule derivations (see Text S2 in Supporting Information S1):

$$\frac{d\tau_{1,i}}{dt} = \left(1 + \frac{\mu}{2c_s}\frac{\partial V_{1,i}}{\partial \tau_{1,i}}\right)^{-1}\left[-\sum_{j=1}^{N} k_{ij}^{s1} V_{1,j} + \dot{\tau}_{1,i} - \frac{\mu}{2c_s}\left(\frac{\partial V_{1,i}}{\partial \sigma_i}\frac{d\sigma_i}{dt} + \frac{\partial V_{1,i}}{\partial \psi_i}\frac{d\psi_i}{dt}\right)\right] \tag{5}$$

$$\frac{d\tau_{2,i}}{dt} = \left(1 + \frac{\mu}{2c_s}\frac{\partial V_{2,i}}{\partial \tau_{2,i}}\right)^{-1}\left[-\sum_{j=1}^{N} k_{ij}^{s2} V_{2,j} + \dot{\tau}_{2,i} - \frac{\mu}{2c_s}\left(\frac{\partial V_{2,i}}{\partial \sigma_i}\frac{d\sigma_i}{dt} + \frac{\partial V_{2,i}}{\partial \psi_i}\frac{d\psi_i}{dt}\right)\right] \tag{6}$$

$$\frac{d\sigma_i}{dt} = \sum_{j=1}^{N} k_{ij}^{N} V_j + \dot{\sigma}_i \tag{7}$$

$$\frac{d\psi_i}{dt} = \frac{b}{d_c}\left[V_0 \exp\left(\frac{f_0 - \psi_i}{b}\right) - V_i\right] \tag{8}$$

The system includes both shear components (strike-slip $\tau_{1,i}$, dip-slip $\tau_{2,i}$) and the normal stress σ_i at i-th element, as well as the regularized state variable ψ_i. These are assembled into a vector of size $4N$:

$$\frac{dy}{dt} = f(y) \tag{9}$$

$$y = \left(\psi_1, \ldots, \psi_N, \tau_{1,1}, \ldots, \tau_{1,N}, \tau_{2,1}, \ldots, \tau_{2,N}, \sigma_1, \ldots, \sigma_N\right) \tag{10}$$

We solve this system using an adaptive 5th-order Dormand-Prince Runge-Kutta integrator (Press et al., 2007). After each iteration, we update the slip velocity and associated fault variables:

$$\tau = \sqrt{\tau_1^2 + \tau_2^2} \tag{11}$$

$$V_1 = 2V_0 \cdot \exp\left(-\frac{\psi}{a}\right) \cdot \sinh\left(\frac{\tau_1}{a\,\sigma}\right) \tag{12}$$

$$V_2 = 2V_0 \cdot \exp\left(-\frac{\psi}{a}\right) \cdot \sinh\left(\frac{\tau_2}{a\,\sigma}\right) \tag{13}$$

$$V = \sqrt{V_1^2 + V_2^2} \tag{14}$$

$$\text{rake} = \arctan\left(\frac{\tau_2}{\tau_1}\right)$$

The schematic in Figure 3d shows the decomposition of both shear stress and slip vector components and the definition of the rake angle on a triangular fault element. This geometric framework is used to resolve local slip into orthogonal directions and compute traction evolution consistently.

The adaptive time step h_{n+1} is selected based on local error estimates:

$$h_{n+1} = S\, h_n \left(\frac{\epsilon_0}{\epsilon_k}\right)^{0.2} \tag{15}$$

$$\epsilon_k = \max\left(\left|\frac{y_{n+1} - y_{n+1}^*}{y_{n+1}}\right|\right) \tag{16}$$

where $S = 0.9$ is a safety factor, and $\epsilon_0 = 10^{-4}$ is the tolerance. If $\epsilon_0/\epsilon_k < 1$, the step is accepted and scaled accordingly. Otherwise, the step is rejected and recomputed with a reduced h_{n+1}, ensuring stability and accuracy. The y_{n+1}^* is the result of the fifth order computation and y_{n+1} is the result of the fourth order computation.

To ensure numerical resolution of rupture, we monitor two key physical length scales. The first is the process zone size Λ, given by:

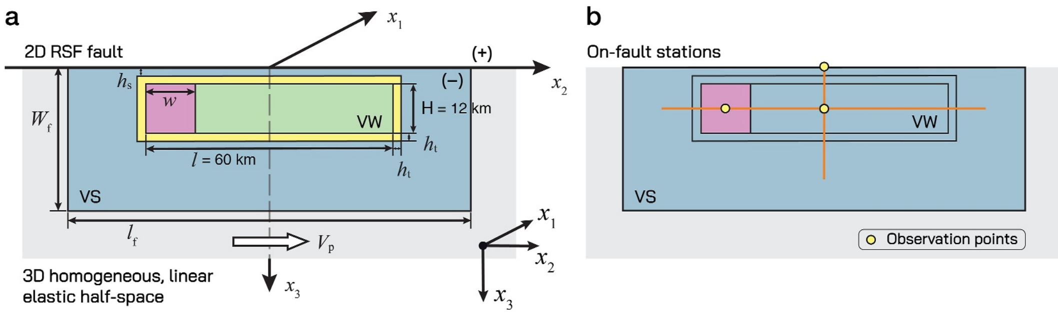

Figure 4. BP5-QD benchmark setup (Jiang et al., 2022). (a) Schematic of the vertical strike-slip fault in a 3-D homogeneous, linear-elastic half-space. The fault hosts a velocity-weakening (VW) patch (light green) of length $l = 60$ km and depth $H = 12$ km, bounded by velocity-strengthening (VS) regions (blue). A rectangular nucleation patch (purple, width w) centered near the left edge of the VW zone is slightly overstressed to trigger the first earthquake. (b) Locations of three on-fault observational points (yellow circles) used for time-series diagnostics: (1) a *nucleation-zone* receiver at ($x_1 = 0$, $x_2 = -24$ km, $x_3 = 10$ km) inside the purple patch; (2) a *mid-VW* receiver at ($x_1 = 0$, $x_2 = 0$, $x_3 = 10$ km), the center of the VW region; and (3) a *surface* receiver at ($x_1 = 0$, $x_2 = 0$, $x_3 = 0$) directly above the mid-VW station. Orange lines denote the along-strike and along-dip profiles used for slip and stress diagnostics.

$$\Lambda = C\frac{\mu\, d_c}{b\, \sigma_n} \tag{17}$$

which characterizes the breakdown zone near the rupture front. Here, C is a dimensionless constant, typically taken as $9\pi/32$ Lapusta and Liu (2009). Three-dimensional dynamic rupture simulations by Day et al. (2005) demonstrated that a resolution of at least $\Lambda/\Delta x = 3$–5 is required to adequately capture the physical features of rupture propagation. Figure 3b shows a snapshot of the resulting shear stress field, with localized gradients near the VW/VS interface consistent with this scale. The second critical length scale is the nucleation size h^*, defined as (T. Chen & Lapusta, 2009; Rice & Ruina, 1983; Rubin & Ampuero, 2005):

$$h^* = \frac{\pi}{2}\cdot\frac{\mu\, b\, d_c}{(b-a)^2\,\sigma_n} \tag{18}$$

This scale governs the minimum size of the VW patch required for spontaneous nucleation. A snapshot of fault slip rate, shown in Figure 3c (log-scale), illustrates the nucleation and arrest behavior constrained by this scale. These two length scales (Λ, h^*) must be appropriately resolved by the mesh to ensure physically meaningful and numerically stable earthquake cycle simulations.

4. Validation Against Benchmarks

To evaluate the accuracy, stability, and performance of PyQuake3D, we validate the code against Benchmark Problem 5 (BP5-QD) from the SEAS (Sequences of Earthquakes and Aseismic Slip) benchmark suite (Jiang et al., 2022). This benchmark is a standard quasi-dynamic problem designed to test the ability of a simulator to reproduce the full earthquake cycle, including nucleation, dynamic rupture propagation, postseismic slip, and interseismic loading.

Figure 4a illustrates the fault configuration for BP5-QD. The model consists of a 2D vertical strike-slip fault embedded in a 3D homogeneous, linear elastic half-space. The fault is governed by rate-and-state friction with the aging law. A velocity-weakening (VW) region ($l = 60$ km along-strike, $H = 12$ km in depth) is surrounded by velocity-strengthening (VS) material. The nucleation of the first event is triggered by applying a slight overstress in a small patch within the VW zone. The entire domain is loaded at a constant plate rate V_p through imposed far-field boundary conditions. The complete set of material and friction parameters is listed in Table 1.

All simulations are run at 1 km resolution using 9,200 triangular elements and 21,500 time steps. A set of on-fault observation stations (Figure 4b) are used to record slip rate and stress evolution across the fault. These include three monitoring points—one located at the center of the VW patch, one point at the free surface, and one within

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Table 1
Frictional and Material Parameters for the BP5-QD Benchmark (Jiang et al., 2022)

Parameter	Symbol	Value	Unit
Reference slip rate	V_0	10^{-6}	m/s
Initial slip rate	V_{init}	10^{-9}	m/s
Effective normal stress	$\bar{\sigma}_n$	25	MPa
Reference friction coefficient	f_0	0.6	–
Frictional parameters (VW)	a, b	0.04, 0.03	–
Frictional parameters (VS)	a, b	0.004, 0.03	–
Characteristic slip distance	d_c	0.14, 0.13[a]	m
Density	ρ	2,670	kg/m^3
Shear wave speed	c_s	3.646	km/s
Poisson's ratio	ν	0.25	

[a]The value used in the prescribed nucleation zone.

the prescribed nucleation zone. PyQuake3D completes the entire benchmark simulation in under 50 min on a standard workstation, averaging 8.5 s per 100 time steps. This result demonstrates both computational efficiency and numerical robustness across multiple seismic cycles.

Figure 5 shows the cumulative along-strike slip distributions for five earthquake cycles, comparing PyQuake3D with results from FDRA (Segall & Bradley, 2012) and BICyclE (Lapusta & Liu, 2009). All codes exhibit similar rupture patterns, with bilateral rupture propagation, repeatable nucleation near the center of the VW patch, and gradual postseismic slip in the surrounding VS zones. Slight variations in rupture initiation and termination points across cycles are consistent with previous benchmark studies and likely reflect differences in numerical resolution and loading accumulation. The results confirm that PyQuake3D captures the spatial evolution of coseismic and interseismic slip with consistency and fidelity.

To further evaluate the accuracy of PyQuake3D, we examine slip rate and shear stress histories recorded at the central observation station located at ($x_1 = x_2 = 0$ km, $x_3 = 10$ km), corresponding to the center of the VW region. Figure 6 compares the time series of (a) maximum slip rate across the fault, (b) slip rate at the central receiver, and (c) corresponding shear stress evolution. PyQuake3D's results are tightly aligned with those of other benchmarked simulators. All codes reproduce the expected sequence of loading, nucleation, rapid rupture, and stress drop with consistent recurrence intervals (~200 years). Minor discrepancies in peak slip rate or stress amplitude are within the range reported in the SEAS intercomparison study and do not affect the qualitative or quantitative agreement. We further compare the waveforms of different earthquake events in Text S3 in Supporting Information S1, to demonstrate the code's capability to accurately reproduce the co-seismic slip characteristics of fault surface (Figures S2 and S3 in Supporting Information S1).

Overall, these results confirm that PyQuake3D reproduces the key physical features of BP5-QD, including earthquake recurrence, rupture dynamics, stress evolution, and interseismic slip accumulation. Its consistency with established codes and its efficiency on standard hardware demonstrate that PyQuake3D is a reliable and practical tool for 3D quasi-dynamic earthquake cycle simulations.

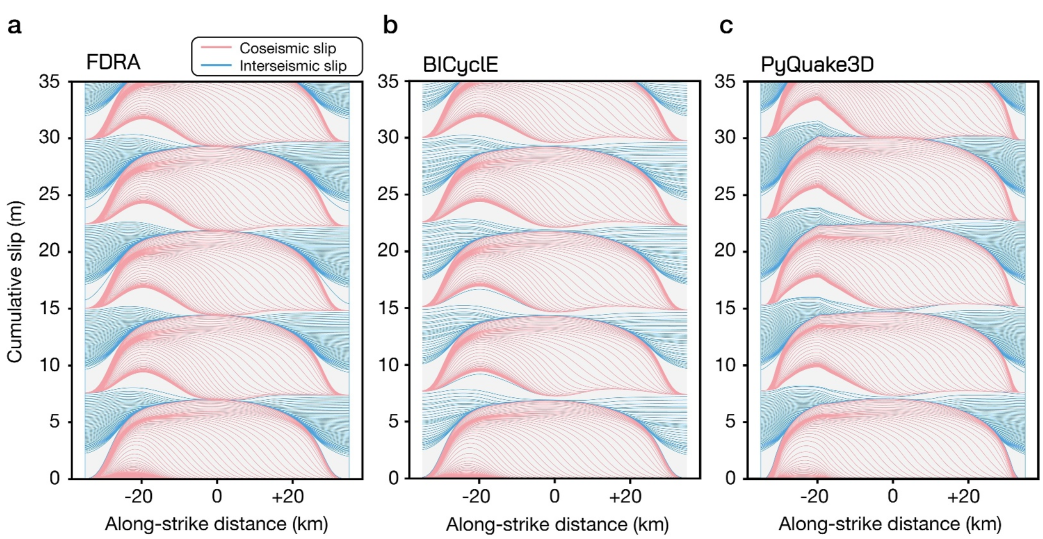

Figure 5. Cumulative slip profiles along strike over five earthquake cycles for three different simulators: (a) FDRA, (b) BICyclE, and (c) PyQuake3D. Coseismic (red) and interseismic (blue) slip profiles reveal consistent rupture nucleation, propagation, and arrest across all codes.

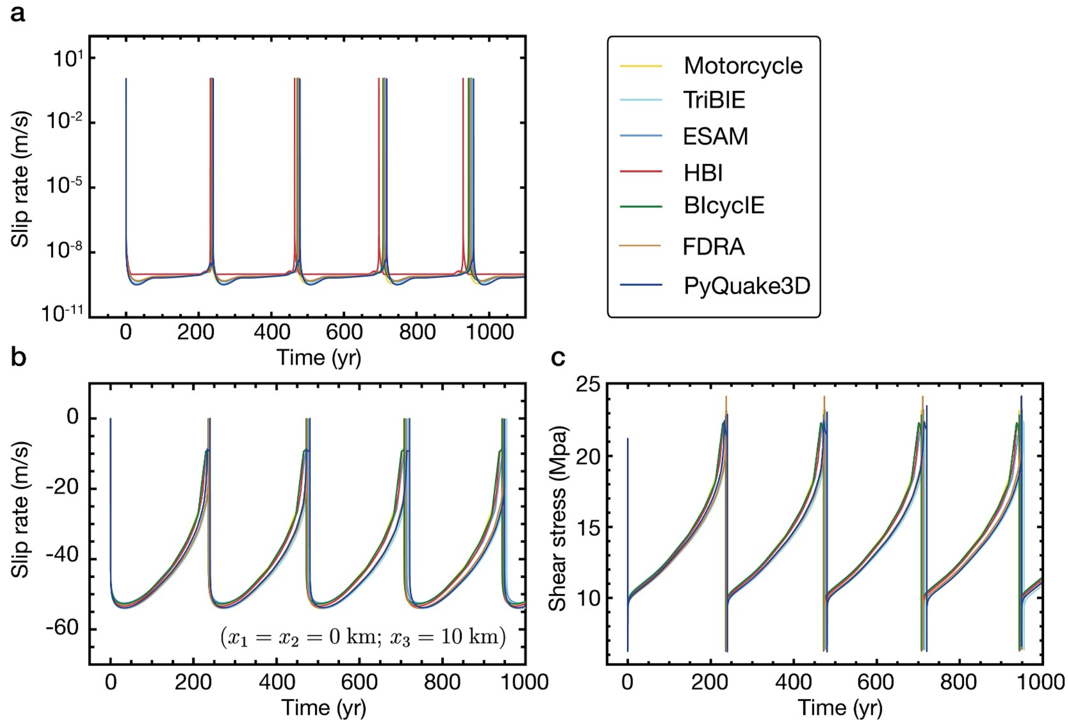

Figure 6. Comparison of benchmark results across different SEAS simulators, including PyQuake3D. (a) Maximum slip rate across the fault over time. (b) Slip rate at the central on-fault receiver located at $(x_1 = x_2 = 0, x_3 = 10)$ km. (c) Shear stress at the same location. PyQuake3D results are in close agreement with the reference solutions across all metrics.

5. Applications of PyQuake3D to Complex Fault System

To demonstrate the flexibility and capabilities of PyQuake3D in modeling diverse faulting scenarios, we present three illustrative examples involving: (a) a planar fault with spatially heterogeneous frictional properties, (b) a geologically realistic multi-segment fault system based on the East Anatolian Fault Zone (EAFZ), and (c) a triaxial laboratory experiment involving a saw-cut sample with frictional weakening. These cases highlight the ability of PyQuake3D to reproduce a wide range of slip behaviors, from microseismicity to large earthquakes, and to capture both the spatial complexity and temporal variability of fault processes. Although none of the simulations are designed to reproduce specific natural events, they serve to validate the capacity of the code to resolve spontaneous rupture nucleation, aseismic slip transients, and the long-term evolution of complex fault systems under varying frictional and geometric conditions.

For completeness, model parameters for all case studies are listed in Table 2. In both the heterogeneous fault (HF) and East Anatolian Fault Zone (EAFZ) models, the initial shear stress τ_0 is set based on the steady-state solution of the rate-and-state friction law (see Text S3 in Supporting Information S1). In contrast, the triaxial laboratory experiment is designed to mimic true loading conditions, with shear stress gradually increased to initiate slip, consistent with laboratory protocols. We emphasize that these simulations are not intended to reproduce specific historical earthquakes, but rather to demonstrate the ability of PyQuake3D to model physically consistent rupture sequences in fault systems with realistic geometric and frictional complexity.

5.1. Planar Fault With Frictional Heterogeneity

Frictional heterogeneity—spatial variations in frictional properties along a fault—plays a fundamental role in controlling earthquake initiation, propagation, and termination (Jiang & Lapusta, 2016; Tinti et al., 2005; Yabe & Ide, 2018). These heterogeneities arise from variations in lithology, fault zone damage, gouge composition, and fluid pressure, and they introduce complexity into fault slip behaviors that homogeneous models cannot capture. Previous studies (Jiang & Lapusta, 2016, 2017) have introduced velocity-weakening (VW) patches embedded within velocity-strengthening (VS) backgrounds to simulate microseismic swarms and explore their connection to

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Table 2
Initial Parameters for the Heterogeneous Fault (HF), East Anatolian Fault Zone (EAFZ), and Laboratory (Lab) Experiment Models

Parameter	Symbol	HF model	EAFZ model	Lab model	Unit
Effective normal stress	$\bar{\sigma}_n$	50	20	70	MPa
Frictional parameters (VW)	a, b	0.0185, 0.024	0.04, 0.03	0.0185, 0.024	–
Frictional asperities (VW)	a, b	0.01, 0.025	–	–	–
Frictional parameters (VS)	a, b	0.0185, 0.001	0.004, 0.03	0.0185, 0.001	–
Characteristic slip	d_c	0.015	0.12	1×10^{-7}	m
Plate loading rate	V_p	1×10^{-9}	1×10^{-9}	3×10^{-8}	m/s
Rake angle	*rake*	0	0	−90	Degree
Cell size	Δx	120	1,000	0.0005	m
Number of cells	N	241,446	41,800	14,336	–
Critical nucleation size	h^*	12,661, 1,773	11,149	0.056	m
Cohesive zone	Λ	374, 359	4,710	0.00168	m

mainshock dynamics. Following these approaches, we design a model based on the BP5 benchmark introduced in Section 4, incorporating frictional heterogeneities to assess the capability of PyQuake3D to simulate a broad spectrum of fault slip behaviors (Figure 7).

We performed a 200-year simulation of a frictionally heterogeneous fault model using over 350,000 time steps (Figure 8). Computations were carried out in parallel using MPI on 14 cores of a 12th Gen Intel® Core™ i9-12900K processor, requiring approximately 10 days. The model contains more than 240,000 boundary elements and consumed over 60 GB of memory during execution. With a uniform cell size of 120 m, the critical nucleation size and cohesive zone length within the VW patches are approximately 1,773 and 374 m, respectively, allowing accurate simulation of events down to M_w 4.6 (Figure 7b).

The model captures nucleation, dynamic rupture propagation, postseismic slip, aftershocks, interseismic creep, and slow-slip events with high spatiotemporal resolution (Figure 8g). Rupture initiates at 20 km depth due to aseismic slip acceleration (Movie S1), followed by bilateral propagation of a large earthquake (M_w 7.0; Figure 8b). Subsequent afterslip propagates into adjacent deep regions, accompanied by localized aftershocks (Figure 8c). During the interseismic period, the fault remains mostly locked, with intermittent microseismicity and slow strain accumulation (Figure 8d). Recurrent microseismic swarms shrink the locked zone (Figures 8g and 8h), eventually triggering the next nucleation event (Figure 8f). These results are consistent with prior findings from Jiang and Lapusta (2016).

PyQuake3D successfully captures a broad range of fault behaviors, including the progressive buildup and abrupt release of shear stress (Figure 8h); aseismic creep and afterslip; episodic slow-slip events without seismic

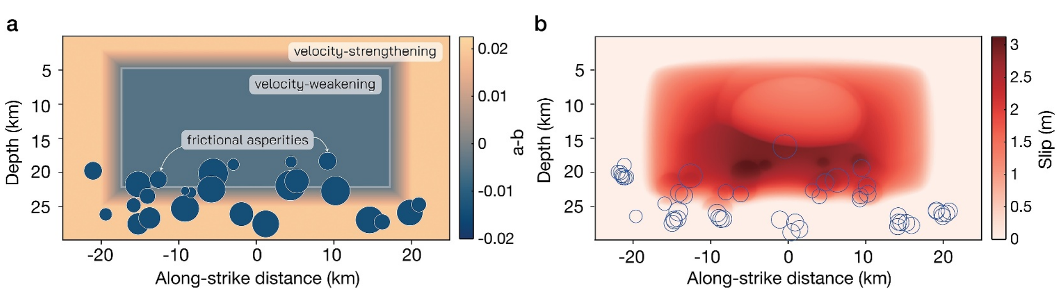

Figure 7. Frictional heterogeneity and rupture behavior in the fault model. (a) Initial distribution of the friction parameter $a - b$, with a central velocity-weakening region bounded by a velocity-strengthening margin. Dark blue circles mark strong velocity-weakening patches representing small-scale asperities. (b) Coseismic slip distribution during the first dynamic rupture, followed by the first slow slip event. Blue circles indicate all simulated microseismic events over the 200-year period.

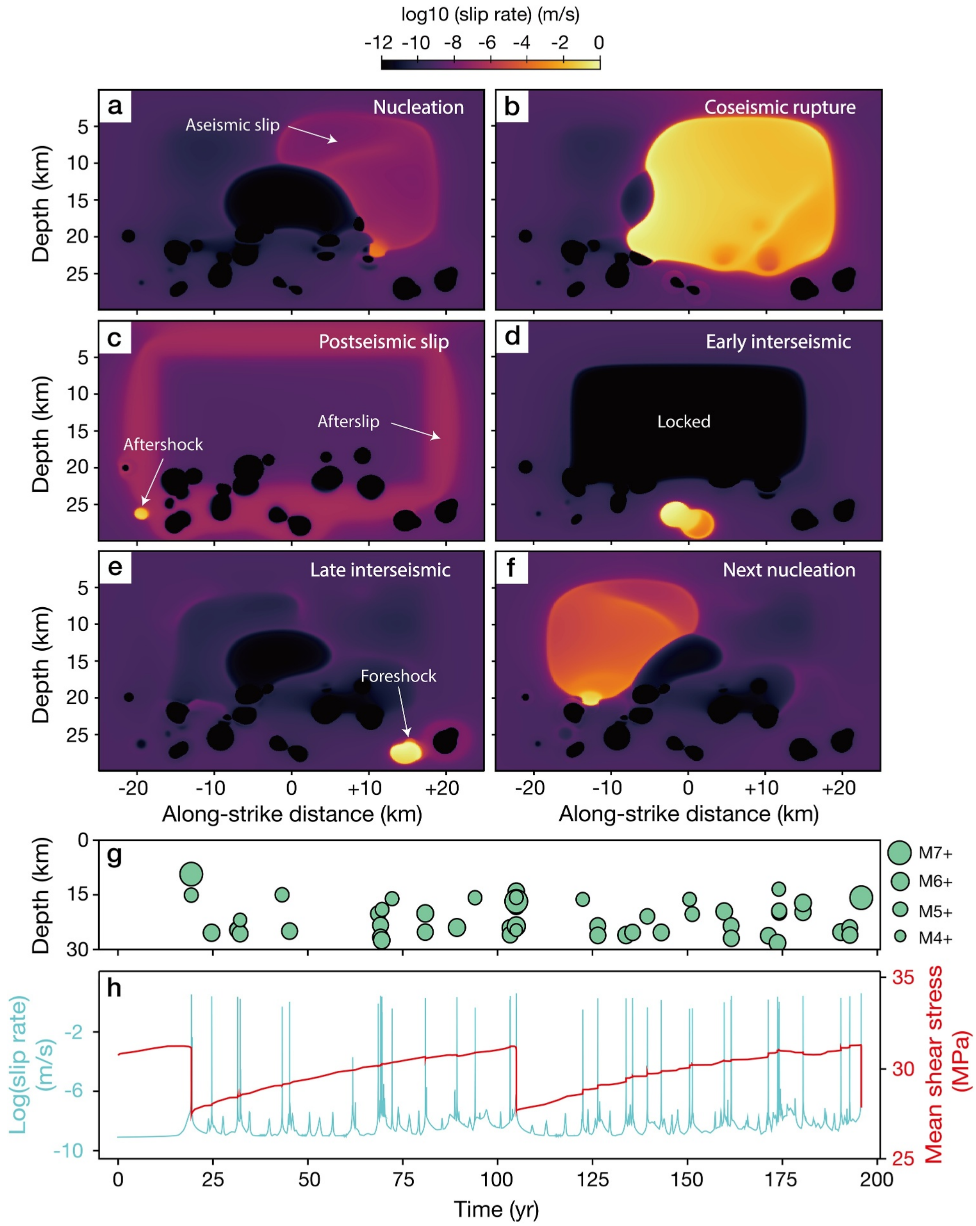

Figure 8. Spatiotemporal variations in slip rate and seismicity in the frictionally heterogeneous model. Panels (a–f) show slip rate evolution during key phases of the seismic cycle. (g) Spatiotemporal variations in event locations and magnitudes. (h) Time evolution of maximum slip rate and average shear stress.

radiation (Movie S1); and recurring microseismic activity surrounding major events. These results highlight the ability of PyQuake3D to model multi-scale interactions between seismic and aseismic slip in geologically motivated settings.

5.2. East Anatolian Fault Zone

The 2023 Mw 7.8 Kahramanmaraş earthquake ruptured a large portion of the East Anatolian Fault Zone (EAFZ), a major left-lateral strike-slip fault system separating the Anatolian and Arabian plates (K. Chen et al., 2024; Dal Zilio & Ampuero, 2023; Wang et al., 2023). Informed by kinematic rupture models (Ren et al., 2024) and high-resolution surface traces extracted from remote sensing (Reitman et al., 2023), we construct a 3D, non-planar fault geometry incorporating the main fault and two branching segments (Figure 9).

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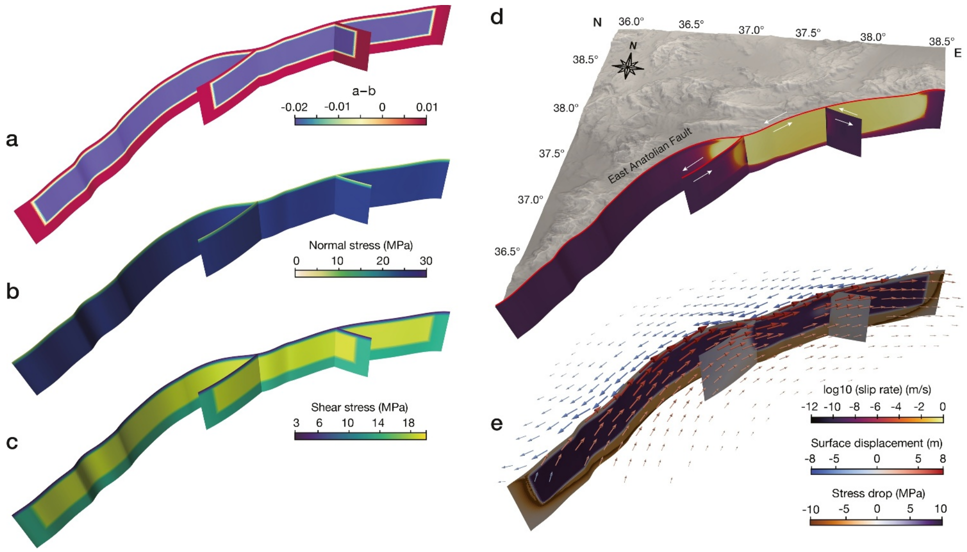

Figure 9. Three-dimensional visualization of the East Anatolian Fault and initial friction and stress configuration. (a) The spatial distribution of the rate-and-state friction parameter (a–b). (b) and (c) are the initial normal stress and shear stress distributions on the fault plane. (d) The slip rate for the first event and topographic map. (e) The stress drop and surface displacement at the end of the first event.

To isolate the effects of complex fault geometry, we impose spatially uniform rate-and-state frictional parameters (Figures 9a–9c). Our aim is not to reproduce the 2023 earthquake in detail, but rather to assess the capability of the model to capture realistic earthquake cycle behavior on a geologically constrained fault structure. The resulting coseismic slip and surface displacement patterns for the first major rupture event (Figures 9d and 9e) show qualitative agreement with those inferred from geodetic and remote sensing observations (Ma et al., 2024). The simulation spans 700 years and includes over 200,000 time steps.

Figure 10 illustrates the long-term evolution of seismic activity. The simulation captures multiple stages of the earthquake cycle: interseismic stress accumulation, nucleation, dynamic rupture, self-healing, postseismic afterslip, and aseismic creep (Figure 10a). The first large event (Mw 8.3) nucleates after 76.8 years of stress accumulation and propagates bilaterally, rupturing the entire main fault and partially entering the western fault branch. This rupture is followed by afterslip along the rupture edges and gradual stress relaxation over a decade (Movie S2).

During the subsequent interseismic period, fault locking resumes, accompanied by local stress accumulation. Two Mw 7.3 events rupture the western fault segment independently of the main fault, whereas multiple slow-slip events (Mw 6.5–6.9) occur on the eastern branch. A particularly interesting sequence occurs at 403.9 years, when a Mw 8.0 earthquake nucleates at the southern tip of the main fault and halts near the branch junction. Just 53 min later, a second Mw 8.0 event nucleates at the edge of the previous rupture and completes the rupture of the entire main fault. This cascade-like sequence demonstrates the potential for delayed triggering and complex stress interactions across fault segments (Figures 10a and 10b). In the final centuries of the simulation, a Mw 7.4 event and several slow-slip episodes occur on the branches. At year 699, another Mw 8.2 event initiates in the southern section and ruptures the entire fault system, illustrating that structural barriers can be overcome under favorable stress conditions.

The co-seismic moment rate functions (Figure 10c) reveal diverse rupture dynamics. Event 1 (Mw 8.3) displays a broad and high-amplitude moment rate curve, characteristic of large, sustained ruptures. In contrast, Event 4 (Mw 8.0) has a short duration and rapid moment release, reflecting a fast rupture arrested by a geometric barrier. Event 6 (Mw 7.4) exhibits multiple peaks, suggesting a complex rupture path with sub-event clustering. These differences underscore the sensitivity of rupture dynamics to prior loading history, stress heterogeneity, and 3D fault geometry (Movie S2).

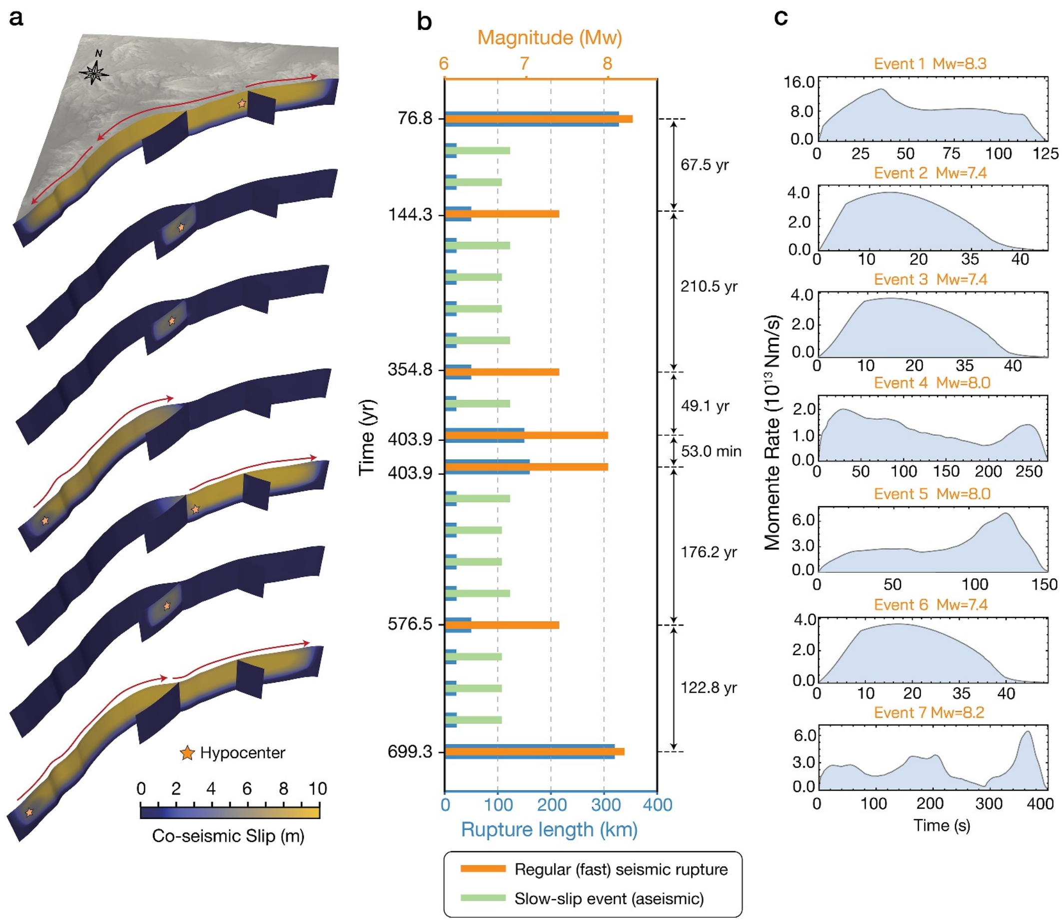

Figure 10. Evolution of seismic activity over a 700-year period of the East Anatolian Fault. (a) Coseismic slips for different regular earthquake events. (b) Statistics of time, earthquake magnitude, and rupture length for all events. (c) The co-seismic moment rate functions for regular earthquakes.

5.3. Modeling Laboratory-Scale Stick-Slip Experiments

Understanding the nucleation and transition from aseismic to seismic slip in frictional interfaces is central to both laboratory seismology and natural fault modeling (e.g., Dublanchet et al., 2024; Mclaskey & Yamashita, 2017). High-resolution triaxial experiments have provided key insights into precursory aseismic slip and the localization of rupture nucleation (e.g., Acosta et al., 2019). Furthermore, recent optical-fiber distributed strain sensing (DSS) has revealed that the bulk sample deforms heterogeneously during the preparatory phase: volumetric strain rates accelerate before macroscopic failure and localize along the emergent rupture plane (Bianchi, Selvadurai, Dal Zilio, Salazar Vásquez, et al., 2024). To test the applicability of PyQuake3D at the millimeter-to-centimeter scale, we simulate a laboratory-inspired stick-slip system using a circular, inclined fault embedded in an elastic cylinder and subject to true triaxial loading (Figure 11).

The numerical setup mimics a cylindrical granite sample with an inclined fault plane at 30° to the axial loading direction, representative of experimental studies such as those by Bianchi, Selvadurai, Dal Zilio, Rast, et al. (2024). The fault surface is discretized into over 14,300 triangular elements with millimeter-scale resolution (Figure 11b). To replicate stress heterogeneity due to fault roughness and loading irregularities, we impose a spatially heterogeneous normal stress field (Figure 11c), whereas the shear stress evolves dynamically in response to imposed displacement at the sample boundaries.

In addition to on-fault dynamics, PyQuake3D allows for the simulation of volumetric strain fields within the host rock. Figure 11d shows the spatial distribution of circumferential and Cartesian strain components post onset of yielding. The results reveal highly localized strain accumulation along the fault plane, with dominant extensional

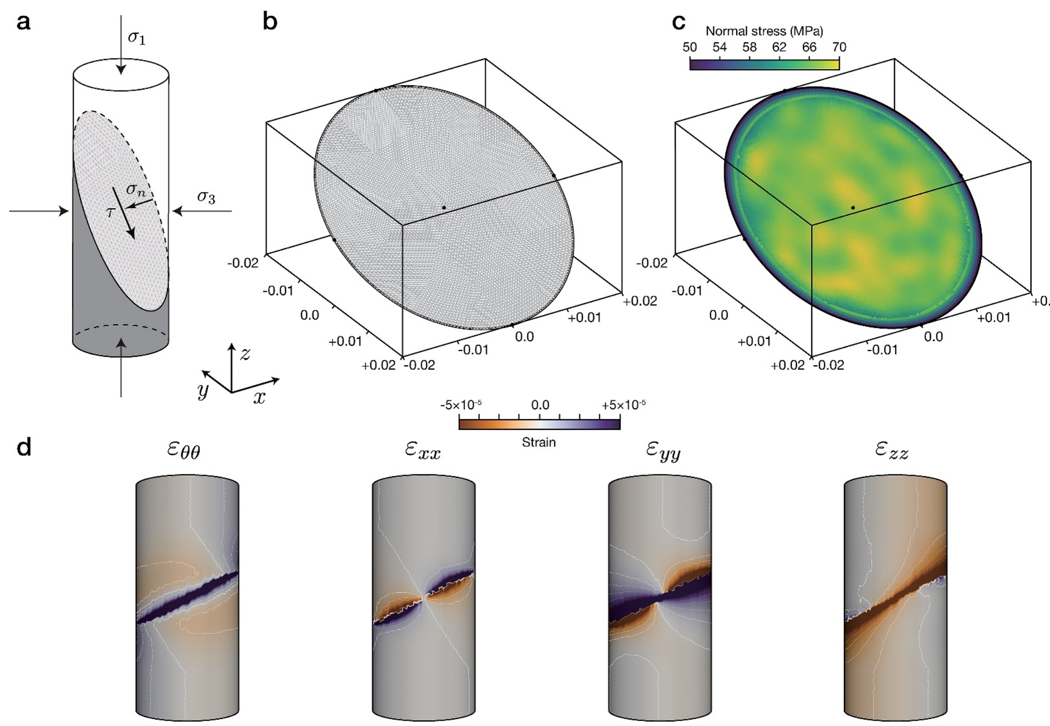

Figure 11. Simulation setup for the laboratory-scale stick-slip model. (a) Schematic of the triaxial loading configuration on a saw-cut cylindrical sample with inclined fault plane. Axial stress σ_1 and confining stresses σ_2, σ_3 are applied to control fault-normal and shear tractions. (b) Unstructured triangular mesh discretization of the elliptical fault surface. (c) Imposed normal stress distribution on the fault surface used to mimic loading heterogeneity and roughness. (d) Volumetric strain fields post onset of yielding, showing the spatial distribution of circumferential strain ($\varepsilon_{\theta\theta}$) and Cartesian strain components (ε_{xx}, ε_{yy}, ε_{zz}). White lines indicate the projected contours of strain.

bands ($\varepsilon_{\theta\theta} < 0$, $\varepsilon_{yy} < 0$) bounded by compressive regions. Axial strain components (ε_{xx}, ε_{zz}) exhibit compressive concentrations at the rupture tips, highlighting the asymmetric loading response of the sample. These patterns are qualitatively consistent with the distributed strain fields recorded in triaxial deformation experiments using Distributed Optical Fiber Sensing, which reveal a progressive transition from distributed compaction to localized shear, enhanced by crack heterogeneity, prior to macroscopic failure under confined conditions (e.g., Miao, Zang, et al., 2025; Miao, Zhang, et al., 2025).

We simulate over 700 s of fault evolution, corresponding to multiple stick-slip cycles under constant boundary loading (Figure 12a). Each cycle begins with aseismic slip and progressive shear stress accumulation, followed by abrupt stress drop and rapid slip acceleration during the dynamic phase. This behavior is consistent with laboratory observations showing that rupture nucleation is often preceded by slow aseismic creep and increasing slip rates (Bianchi, Selvadurai, Dal Zilio, Salazar Vásquez, et al., 2024). The evolution of maximum slip rate and average shear stress highlights the cyclic nature of loading and failure.

Snapshots of slip rate and shear stress during the nucleation and rupture process (Figures 12b and 12c) illustrate the spatiotemporal complexity of failure. Rupture initiates from localized asperities with slightly elevated stress, and accelerates toward full dynamic failure across the fault surface. As in laboratory experiments, precursory slip is observed to concentrate in a limited region before propagating outward during fast slip. Heterogeneous stress patterns persist throughout the cycle due to the imposed roughness and stress feedbacks from prior events. These results demonstrate the ability of PyQuake3D to simulate spontaneous nucleation, slow slip transients, and rupture dynamics in laboratory-scale systems. The spatial and temporal evolution of slip in our simulations qualitatively mirrors laboratory observations of aseismic fault slip preceding dynamic rupture, as revealed by high-resolution kinematic inversions during nucleation phases (Dublanchet et al., 2024). These experimental insights underscore the importance of quasi-static slip transients in controlling rupture onset and support the

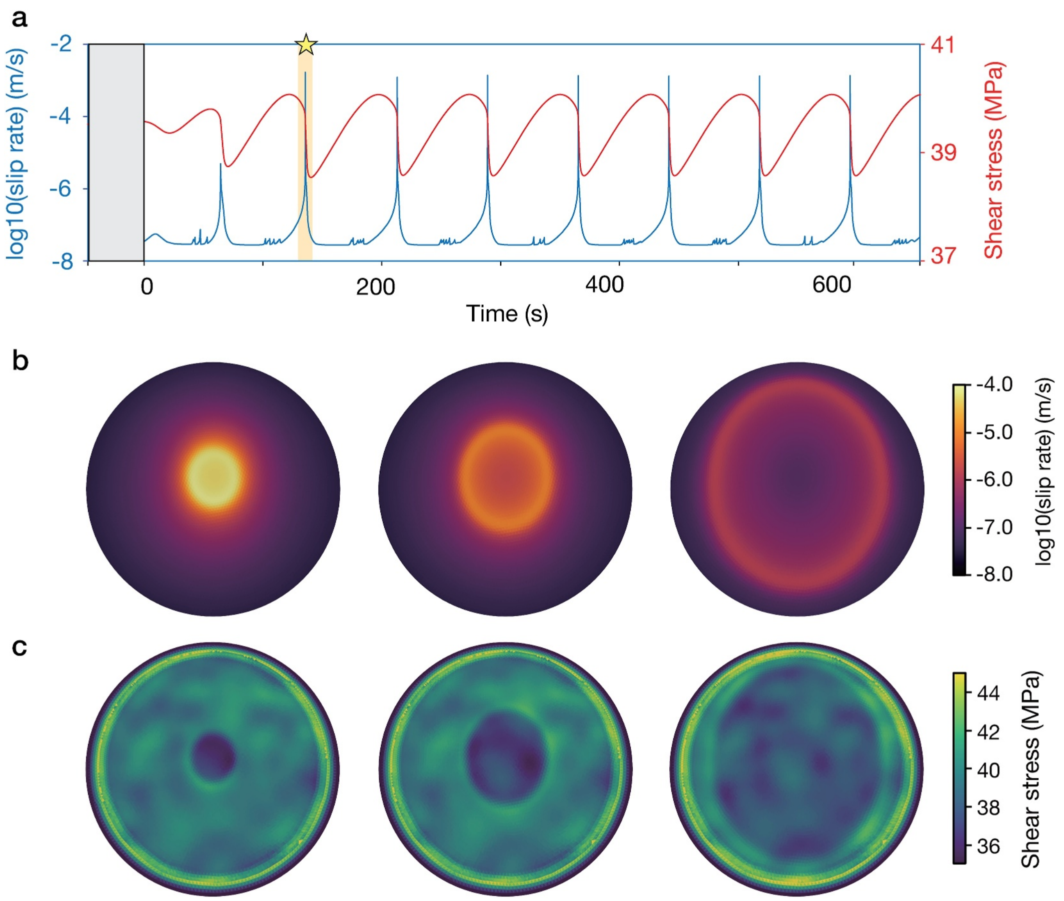

Figure 12. Stick-slip cycles and rupture evolution in the laboratory-scale model. (a) Time series of maximum slip rate (blue) and average shear stress (red) showing multiple recurring stick-slip events. (b) Snapshots of log slip rate at three stages of the rupture process for the event marked by a star in (a). (c) Corresponding snapshots of shear stress evolution, highlighting spatial heterogeneity in stress drop and fault weakening.

relevance of rate-and-state frameworks and boundary integral methods for bridging laboratory-scale processes with natural fault behavior.

6. Discussion and Conclusion

6.1. Technical Advances in PyQuake3D

PyQuake3D addresses key computational and modeling challenges that arise in simulating long-term earthquake sequences on geometrically complex faults in three dimensions. Unlike existing codes optimized either for dynamic rupture or for simplified planar settings (e.g., Lapusta & Liu, 2009; Romanet & Ozawa, 2021), PyQuake3D combines scalability, accessibility, and generality in a unified, open-source platform. Its architecture is designed to accommodate a wide range of physical applications—from spontaneous rupture nucleation and propagation to aseismic slip, afterslip, and laboratory-scale fault behavior.

A central innovation is the implementation of a fast hierarchical matrix (H-matrix) solver for the boundary element stiffness matrix. This reduces the computational complexity from $\mathcal{O}(N^2)$ to approximately $\mathcal{O}(N \log N)$, where N is the number of elements. The H-matrix exploits the low-rank structure of elastic interactions and is constructed using adaptive cross approximation (ACA), avoiding the need to assemble or store the full dense matrix. The solver is parallelized using `mpi4py`, with minimal communication overhead, and has been tested on large-scale problems with up to 300,000 triangular elements—an order of magnitude beyond the practical limits of conventional dense approaches. Complementing the CPU-based H-matrix solver, PyQuake3D includes a GPU backend that performs dense matrix-vector multiplications using `CuPy` and CUDA-accelerated routines. This allows for rapid prototyping and efficient simulations on smaller meshes using local GPU resources. Crucially, both backends share a common interface: the same model configuration can be executed interchangeably on CPU or GPU with no change to user inputs, offering flexibility and robustness in research workflows. Such dual-backend architecture is rarely available in community earthquake simulation codes.

PyQuake3D also supports fully unstructured triangular fault meshes, compatible with `Gmsh` (Geuzaine & Remacle, 2009), enabling the modeling of realistic fault geometries including kinks, stepovers, and branched structures. The internal mesh handling is optimized for parallel simulations, incorporating built-in partitioning tools to support distributed H-matrix assembly and solver execution. Frictional behavior is specified through modular Python classes that can be extended to define custom rate-and-state laws, slip-weakening models, or time-dependent parameters. This modular design allows users to explore a wide range of frictional phenomena and test physical hypotheses without modifying the core solver routines.

The numerical integration of the governing ODE system employs a fifth-order Dormand-Prince Runge-Kutta method with adaptive time stepping (Press et al., 2007), which balances accuracy and efficiency across multiple timescales. PyQuake3D also supports the evaluation of mesh-dependent resolution metrics, such as the nucleation size h^* and cohesive zone length Λ, enabling users to assess the physical fidelity of their discretization relative to frictional parameters and stress gradients. This ensures that key features of spontaneous rupture dynamics are resolved consistently across model resolutions.

From a software engineering perspective, PyQuake3D is fully implemented in Python and relies on open-source scientific computing libraries. Simulation outputs are stored in structured `NumPy` and `VTK` formats, enabling seamless integration with machine learning ecosystems such as `PyTorch`, `TensorFlow`, and `JAX`. The modular separation between frictional physics, numerical solvers, and I/O routines facilitates downstream workflows including surrogate modeling, uncertainty quantification, and hybrid physics-ML approaches. PyQuake3D is particularly well suited for coupling with data assimilation techniques, such as ensemble Kalman filtering and physics-informed neural networks (PINNs) (e.g., Fukuda & Barbot, 2025; Fukushima et al., 2025). These capabilities position PyQuake3D as a computational backbone for building real-time, data-driven digital twins of active fault systems, bridging deterministic simulations with data-centric modeling to enable intelligent forecasting and interpretable AI tools for seismic hazard analysis (Dal Zilio et al., 2023).

6.2. Future Developments and Outlook

PyQuake3D establishes a flexible and high-performance foundation for quasi-dynamic earthquake cycle simulations, and ongoing developments aim to expand both its computational capabilities and its physical modeling scope. On the computational side, efforts are underway to implement a hybrid CPU-GPU architecture that combines the scalability of the H-matrix CPU solver with the speed of GPU-accelerated kernels (H. Huang et al., 2012). This hybridization will enable more efficient load balancing, better memory utilization, and support for heterogeneous compute environments, making it possible to simulate fault networks with millions of degrees of freedom. In parallel, ongoing improvements in memory handling, mesh partitioning, and matrix compression are expected to reduce runtime and extend the range of solvable problems, bringing quasi-dynamic modeling closer in scale and fidelity to fully dynamic rupture simulations. Additional enhancements will target modularity and generalization of the solver backends, including dynamic solver switching, support for mixed-precision arithmetic, and integration with sparse and hierarchical matrix libraries beyond the current implementation. These developments will facilitate not only larger simulations but also more efficient sensitivity analyses, ensemble modeling, and workflow automation in high-performance computing (HPC) environments.

On the physical modeling side, PyQuake3D currently supports effective normal stress fields that vary in space but evolve weakly over time. Further time-dependent evolution requires implementation of pore pressure diffusion along the fault plane, enabling one-way coupling between fluid flow and fault strength. This is central to modeling injection-induced seismicity, natural swarms, and permeability-driven instabilities (e.g., Bhattacharya & Viesca, 2019; Cappa et al., 2019; Zhu et al., 2020). Beyond this, shear-induced dilatancy and compaction can be incorporated to link slip-induced porosity changes to transient pore pressure evolution (Segall & Rice, 1995), allowing for simulations of dilatant strengthening or compactional weakening in partially sealed fault zones (e.g., Dal Zilio et al., 2020; Heimisson et al., 2021; Segall et al., 2010).

Poroelastic deformation of the surrounding crust offers a more complete physical representation of solid-fluid interactions. In undrained conditions, poroelastic feedback permit fault slip to generate pore pressure perturbations that affect the effective normal stress field and promote fault-fault interactions (L. Huang et al., 2025). Such mechanisms have been shown to destabilize mildly rate-strengthening faults and generate slow slip pulses (Heimisson et al., 2019). More advanced multiphysics models couple poroelastic deformation, fluid flow, and fault slip dynamically, capturing the co-evolution of stress, strain, and pore pressure (Dal Zilio, Hegyi, et al., 2022; Dal Zilio, Lapusta, et al., 2022). These models provide a unified framework for simulating fault zone hydraulics, crustal permeability evolution, and hydro-mechanical instabilities at multiple scales.

Incorporation of temperature-dependent processes offers multiple avenues for physical extension. On the one hand, temperature-dependent rheologies enable the modeling of long-term lithospheric deformation, postseismic relaxation, and the brittle-ductile transition (e.g., Dal Zilio, Hegyi, et al., 2022; Dal Zilio, Lapusta, et al., 2022; Lambert & Barbot, 2016; Mallick et al., 2022). These extensions remain compatible with the boundary element framework and allow for comparisons and hybridization with continuum-based simulations. On the other hand, thermally activated weakening mechanisms, including flash heating and thermal pressurization, can be implemented to capture rapid reductions in fault strength during dynamic rupture. Flash heating, which occurs at asperity contacts, introduces extreme rate dependence and has been shown to reduce frictional strength at seismic slip rates (Goldsby & Tullis, 2011; Rice, 1999). Thermal pressurization, resulting from shear heating and fluid expansion, can dramatically lower effective normal stress in fluid-saturated faults and promote dynamic weakening (Rice, 2006). More recently, constitutive formulations that describe latent state evolution as a function of slip rate and temperature provide a unified framework for capturing the complex frictional behavior of fault gouge under varying hydrothermal conditions (Barbot, 2023a).

In summary, PyQuake3D offers a technically innovative and extensible foundation for earthquake cycle simulation. Its modular software design, scalable solvers, and emphasis on reproducibility support a wide range of research applications—from high-resolution rupture modeling to emerging multiphysics investigations. Ongoing and planned developments will expand the code's ability to model coupled physical processes, handle large-scale simulations efficiently, and integrate seamlessly with HPC workflows. These advances will ultimately enable more predictive, physically grounded, and data-informed models of fault dynamics across laboratory, crustal, and tectonic scales.

Data Availability Statement

The data supporting this paper are available at Dal Zilio (2025). The `PyQuake3D` source code used in this study and user manual are publicly available at Tang and Dal Zilio (2025). The version of the code hosting ongoing development and future releases of the code is openly accessible on GitHub at https://github.com/Computational-Geophysics/PyQuake3D.

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